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Intellectual Innovations in Georgia (11th-9th Centuries BC)

Excavations at Grakliani Hill

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Abstract

The multilayer archaeological site Grakliani hill is located in one of the main contact zones of Transcaucasia. It shows a very intensive level of communication with other cultures from the 2nd millennium BC till the 3rd century BC and a high level of development of Kartli (*Iberia Caucasica*) society. Two inscriptions made in unknown script (probably a local version of Aramaic script) were discovered in a shrine of the 11th-9th centuries BC. A group of weights and tokens, also of the 11th-9th centuries BC was discovered in the excavations of Grakliani Hill and the satellite site of Tsina Gora. The Achaemenid period remains are also of special interest, including Greek style architecture details.

Keywords

Iberia – Georgia – Grakliani – multilayer site – Aramaic script – inscriptions – weights – tokens

The Grakliani Hill settlement and necropolis, as well as Tsina Gora site (4 km to the South-East from Grakliani Hill) are situated in the central part of the South Caucasus, in Georgia. On the terraces of this hill (fig. 1), numerous pottery sherds, as well as bronze, silver and gold objects, were found. Along with the local materials, there is an impressive collection of Mesopotamian,

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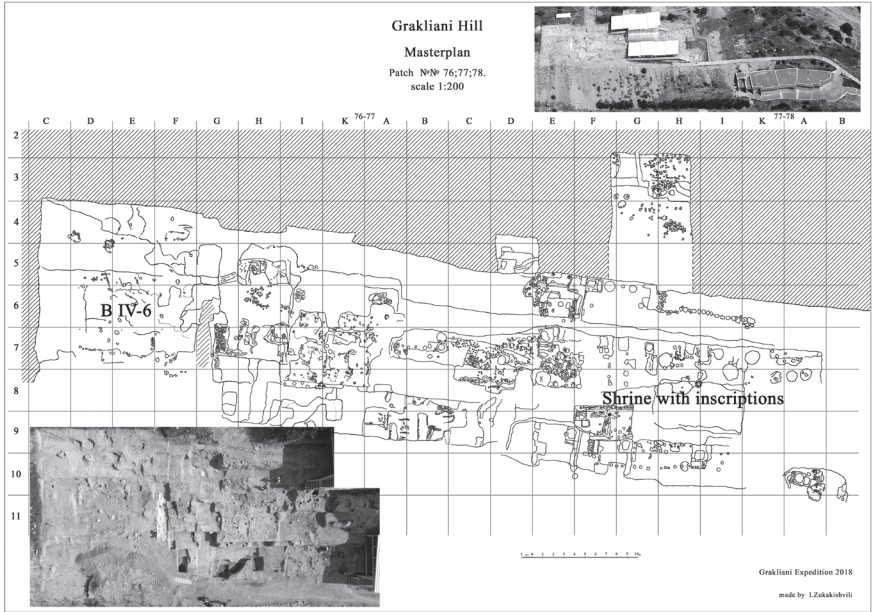


FIGURE 1 Grakliani hill, 3rd and 4th terraces. Masterplan

Urartian, Persian, Greek and Egyptian imports.¹ The materials discovered at Grakliani date to the period from the Paleolithic up to the 4th century AD.

Achaemenid Period

The levels of the Achaemenid period (the second stage of the “Grakliani Culture”) dating to the 5th and 4th centuries BC, are especially well preserved on the 2nd, 3rd and 4th terraces,² including the building IV-6 with a fire altar in the center and a grinding stones on the north wall (fig. 2). In the north-western corner, an iron ingot, as well as a large amount of pottery, were found. A hoard of iron objects was found in the south-eastern corner, under the floor; it is probably a sacrificial deposit. In the north-eastern corner, two *thymiateria* for the holy fire stood on an oven, on which iron spearheads and a knife were also found (fig. 3). As the fire altar dominated the interior of the room and sacrificed objects and vases connected with the cult were found on it, we can assume that

1 Licheli 2009a; 2009b; 2010; 2011; 2013; 2016; 2020; Licheli & Samsonia 2010. The excavations were funded by the National Agency for Cultural Heritage Preservation of Georgia.
2 Litcheli 2017, 41-58; Licheli 2020, 48-57.



FIGURE 2
Building IV-6.
General view



FIGURE 3
Building IV-6.
Oven with iron
objects and two
thymiateria

this building was used as the shrine of a deity of metallurgy. The workshop, where iron tools were produced, was located east from building IV-6, at 20 m distance, on the 3rd terrace. Beside the building IV-6 and the workshop on the 3rd terrace (building III-3), a crucible with copper traces and two kilns for pottery production (11th and 6th centuries BC) were excavated. Despite the fact that silver and gold objects were found in graves (for example, grave 19-4 with horse skeleton and silver jewelry – bracelets, *fibulae*, neckless – similar to those from Vani grave 24³) (fig. 4), there is no evidence of precious metal production

3 Kacharava & Kvirkvelia 2009, 237-316.



FIGURE 4 Grave 19-4 with horse skeleton and silver jewelry, first half of the 4th century BC



FIGURE 5 Greek architectural details from the 5th terrace

at Grakliani.⁴ Another important find was made in the test trench v-1 on the 5th terrace – a Greek architectural fragments of the 5th or 4th centuries BC (fig. 5). The main building of the sanctuary was constructed in the 5th or 4th century BC and burnt in the 30s or 20s of the 4th century BC. This destruction is to be related to Mithrenes' invasion. Mithrenes, ex-commander of the Persian garrison at Sardis, was appointed satrap of Armenia by Alexander the Great in 331 BC and probably conquered eastern Georgia for Alexander.⁵

4 The problem of precious metal production in Georgia, especially of early gold, is the subject of long-time discussion. Cf. "there is no evidence of a gold or silver manufacturing tradition during the Kura-Araxes period in the southern Caucasus. Golden objects are virtually unknown despite gold mining possibly being undertaken in the Kura-Araxes phase, as suggested by the discovery (by the Deutsches Bergbau-Museum of Bochum) of a gold mining site from the 4th Millennium BC in the Sakdrisi area in southern Georgia": Carminati 2014, 172, see also p. 179. For golden objects of the Chalcolithic and Early Bronze Age, see Helwing 2013, 105-135; Hansen 2013, 137-167. For a new chronology see Pollard *et alii* 2013.

5 Licheli 2020, 48-57.



FIGURE 6 Shrine with inscriptions. View from the South

The Cultural System of the Pre-Achaemenid Period: Shrine with Inscriptions

The Achaemenid period construction on the third terrace was preceded by a more ancient shrine. The plaster of its eastern wall was discovered in section F8 on the 3rd terrace (fig. 6). The entire building area (6×2.7 m) was covered by various vessels on the same level. A massive clay construction interpreted as an altar was attached to the western wall of the building. There are three signs ("inscription A") on the north corner of the altar. Another smaller (0.55×0.40 m) clay altar was found in the central part of the building, at 0.5 m distance from the northern wall; it was probably originally attached to northern wall and slipped down later from its original location. The altar was placed on a clay pedestal. On its front another inscription ("inscription B") was found (fig. 7). It is 80 cm long and 7 cm high. The inscription is done freely in clay, the lines are drawn easily. It was obviously not the first attempt of writing and other inscriptions of the same period could be found in the future.

The inscription B is part of a ceramic assemblage dating to the 11th and 10th centuries BC. Radiocarbon analyses of organic remains from the same assemblage made in Miami Beta laboratory, gave a calibrated date range between 1020 and 900 BC. The inscription is not yet read and it is impossible to say anything about the meaning of the text. However some hypotheses are possible. The inscription probably has cultic character as it was carved on the



FIGURE 7 Inscription B

pedestal of an altar. Its script could be a local version of the Aramaic script; in this case it would be one of the earliest cases of its use. The letters of the two inscriptions, A and B, are different from each other. These inscriptions made by local population/priests are unique not only for Georgia, but for the whole Caucasus.

The Economic System: Weights and Tokens

Some unique artefacts contributing to a better understanding of the economic system of Kartli were found during the excavations of Grakliani Hill and its satellite site Tsina Gora. These are weights dating to the 11th-9th centuries BC (fig. 8). The process of weighing is attested as part of the economic activities related to manufacture, transportation and exchange in various Bronze Age societies.⁶ It is usually very difficult to ascertain the units of weight because

6 Pare 1999, 477; Ascalone & Peironel 2000; Michailidou 2010; Bobokhyan 2010, 180; Bobokhyan & Badalyan 2012, 197; Peironel 2015; Alberti *et alii* 2006; Rahmstorf 2006, 9-47.



FIGURE 8
Weights from Grakliani and Tsina Gora
sites

of a lack of artefacts. The objects found in Grakliani and the surrounding area give us such information for eastern Georgia: weight No. 1 is 2 557 g; No. 2 – 918 g; No. 3 – 362 g; No. 4 – 220 g; No. 5 – 15 g; No. 6 – 8 g; The metrological system of the Ancient Near East is well known: the Mesopotamian mina of c. 504 g was initially used in Assyria, but a ‘heavier’ mina, which was double the traditional one (c. 1008 g) seems to have been introduced not later than the beginning of the 9th century BC, becoming the most widely used standard of the empire. The co-occurrence of these two minas is testified by inscriptions on weights, but a clear distinction between a ‘heavy’ (*dannu*) and a ‘light’ (*qallu*) mina is rarely specified in the texts.⁷ We know also about a Near Eastern „light mina” of c. 470 g, which consists of 60 shekel of 7.8 g. The „Levant” system was based on 9.8 g shekel and the „Anatolian” one on c. 11.7 g shekel. A comparison of the weights found on Grakliani with Near Eastern light mina shows that there is no coincidence to each other.

At the same time, one of the weights from Grakliani is trapeze-shaped ($10.2 \times 9.2 \times 2$ cm) and has a hole (diameter 1.2 cm) in the thinner part. Around this hole, it has a circular trace of leather rings and hoops from both sides; these traces show that they were fastened hard. Thus, this weight was hung on a mechanic tool. It might have been a two sided balance, one side of which had a weight of 363 g fastened motionless to it. This means that it was a local weight unit. This hypothesis is confirmed by the find of another weight of 362 g during the excavations on the 4th terrace in 2019. If we compare the Grakliani weights to each other we get the following picture:

⁷ Peyronel 2015, 93-113.

No.	Weight (g)	Multiplicity with local unit (363 g)	Supposed weight according to the local unit	Deviation
1	2557	7	2541	0.7%
2	918	2.5	907.5	1.2%
3	362	1	363	0,3%
4	220	0.6	217.8	1%
5	15	0.04 or 1/25	14.52	3.2%
6	8	1/45	8.07	0.9%

We have to take into account that “one should allow a tolerance of *ca.* $\pm 5\%$ to account for the inherent inaccuracies of ancient balance weight” and also, “we should not forget that the ancient weights are not as precise as ours and they have always fluctuations of some grams”.⁸

During the excavations at Tsina Gora, two clay discs were found, which could be interpreted as tokens used for counting. Tokens were unchangeable, always having only one meaning.⁹ The Tsina Gora tokens are simple clay discs. The first one has eight inequal lines on it, its diameter is 4.2 cm, weight – 13 g. The second one has 12 inequal lines, its diameter is 3 cm, weight – 15 g. The tool used for making lines on these tokens was also found at Tsina Gora. This is an oval stone knife (length 6.6 cm, height 1.7 cm) with a sharp tip.

Conclusions

The economic and cultural systems evidenced by the excavations of Grakliani Hill correspond to a highly developed society (existence of local writing and weighing). However, the excavated area is still too small to give us a clear idea about the economic and cultural situation of the “Grakliani culture” and its interrelations with the neighbors. The excavations should be continued in order to give us more information about this original archaeological culture.

8 Bobokhyan & Badalyan 2012, 197.
9 Schmandt-Besserat 2010, 27-34.

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